

CE6016 - PREFABRICATED STRUCTURES



UNIT 1- INTRODUCTION

- Need for prefabrication
- Advantages
- Materials
- Modular coordination
- Standarization
- Transportation
- Erection

Definition

- The prefabrication which means the production or manufacture of elements which away from the construction site, the elements are called as prefabrication.
- Prefabrication is the practice of assembling components of a structure in a factory or other manufacturing site, and transporting complete assemblies or sub assemblies to the construction site where the structure is to be located.



Process of Prefabrication:

- An example from house building illustrates the process of prefabrication.
- In Conventional method : To transport bricks timber, cement, sand, Steel and construction aggregate etc., to the site, and to construct the house on site from these materials.
- In prefabricated Construction: Only the foundations are constructed in this way while sections of (walls, floors, roof are Prefabricated with window & door frames included) transported to the site lifted into place by a crane and bolted together.



Aim of Prefabrication

- To effect economy in cost
- To improve quality because components can be manufactured under controlled conditions.
- To speed up construction since no curing is necessary
- To use locally available materials

Need for Prefabrication:

- Prefabricated structures are used for sites which are not suitable for normal construction methods such as hilly region, City and also when normal construction materials are not easily available.
- Speedy Construction - No curing period
- Effect Economy
- Improve quality
- Durable structure with less maintenance
- Attractive finishes
- Further expanses easy

Principles of prefabrication techniques:

- Design for prefabrication, preassembly and modular construction.
- To give safety in structural system
- Simplify and standardize connection details.
- Simplify and separate building systems.
- Consider worker safety during deconstruction.
- Minimize building components and materials.
- Select fittings, fasteners, adhesive and sealants that allow for quicker assembly and facilitate the removal of reusable materials.
- Design to accommodate deconstruction logistics.
- Reduce building complexity.
- Design for reusable materials.
- Design for flexibility and adaptability.

Choose Precast Construction method over conventional in method

- Economy in large scale project with high degree of repetition in work construction.
- Special requirement in finishing.
- Consistency in structural quality control.
- Fast speed of construction.
- Constraints in availability of site resources (e.g. materials & Laborites)
- Other space & environmental constraints.

Advantages of prefabrication

- Self-supporting ready-made components are used, so the need for formwork, shuttering and scaffolding is greatly reduced.
- Construction time is reduced.
- Speed up construction because of curing is not necessary.
- Improve the quality as the components can be manufactured under control condition.
- Use locally available materials with required characteristics.
- Minimize of wastage.
- Onsite construction is minimized.
- Requirement of skilled labors such as mason, carpenters, bar benders etc., can be reduced.
- It save manpower.
- Shrinkage of the units which largely eliminated.
- Less expansion joints are needed.
- More accurate & better workmanship.
- Cross section of member can be reduced by use of high strength concrete.

Disadvantages of prefabrication

- Careful handling of prefabricated components such as concrete panels or steel and glass panels is required.
- Attention has to be paid to the strength and corrosion-resistance of the joining of prefabricated sections to avoid failure of the joint.
- Similarly, leaks can form at joints in prefabricated components.
- Transportation costs may be higher for voluminous prefabricated sections than for the materials of which they are made, which can often be packed more compactly.
- Height restrictions under bridges.
- Road transport maximum widths.
- Additional cost of temporary bracing for transportation and/or lifting or permanent framing to support prefabricated assemblies.
- Large prefabricated sections require heavy-duty cranes and precision measurement and handling to place in position.
- Greater Erection equipments are needed.

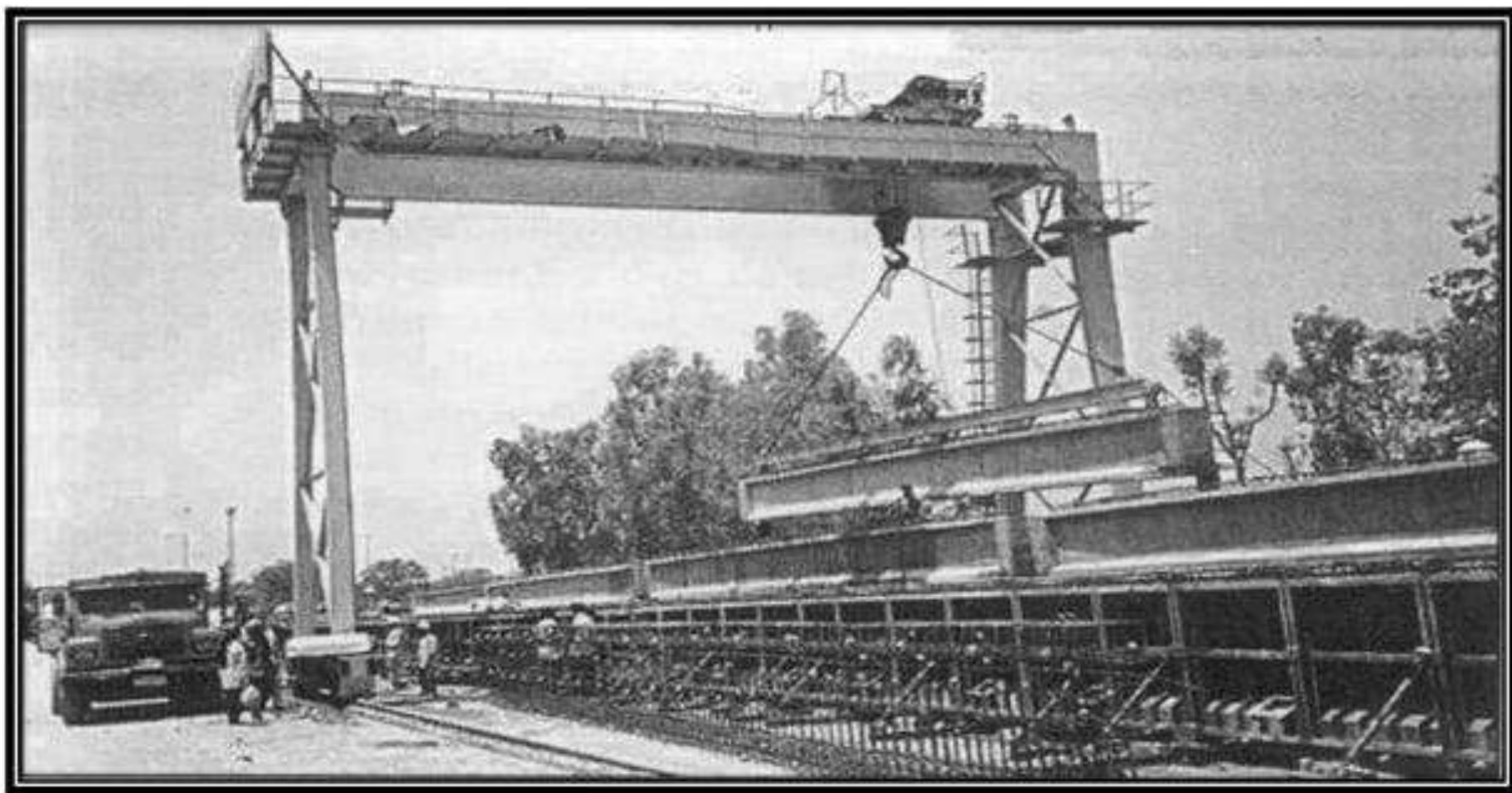
Casting



Curing



Transportation and Erection



Limitations of Prefabrication

- Very Costly
- Uneconomical for smaller projects
- Transportation difficulties
- Safety of Crane and other support system has to carefully considered.
- Difficulties in transportation of vary large components
- Water proofing joints

Uses of Prefabrication

The following are the uses of introducing the prefabrication system:

- Prefabricated components speed up construction time, resulting in lower labor costs;
- Prefabrication allows for year-round construction;
- Work is not affected by weather delays (related to excessive cold, heat, rain, snow, etc.);
- The mechanization used in prefabricated construction ensures precise conformity to building code standards and greater quality assurance;
- There are less wasted materials than in site-built construction;
- There is less theft of material/equipment (and less property damage due to vandalism);
- Materials are protected from exposure to the elements during construction;
- Worker safety and comfort level are higher than in site-built construction;
- Quality control and factory sealing and design can ensure high energy

Materials used for Prefabrication:

- Materials like concrete, steel, treated wood, aluminum, Cellular concrete, Light weight concrete etc...
- Special characteristics while choosing materials
 - Light weight for easy handling and transport and to economic an sections and sizes of foundations.
 - Easy available
 - Economy
 - Easy workability
 - Durability
 - Thermal insulation property
 - Sound insulation

Materials used for Prefabrication

(Cont.,)

Material Properties:

- Quick to assemble
- Cost-effective
- Portable/movable
- Strong
- Waterproof, Moisture proof
- Fire Resistant

Materials used for Prefabrication (Cont.,)

- Prefabricated walls are used in steel, wood, fibre glass, plastic or aluminium materials. These materials are cheaper than regular brick and concrete buildings. ■
- For making low cost houses prefabricated materials like Ferro cement consists of a cement matrix reinforced with a mesh of closely spaced iron rods or wires. In this type of construction the techniques used are simple & quick.
- plastic flooring materials can be quickly assembled and are very durable.
- Prefabricated metal buildings use galvanized steel and galvalume as the chief materials for building.

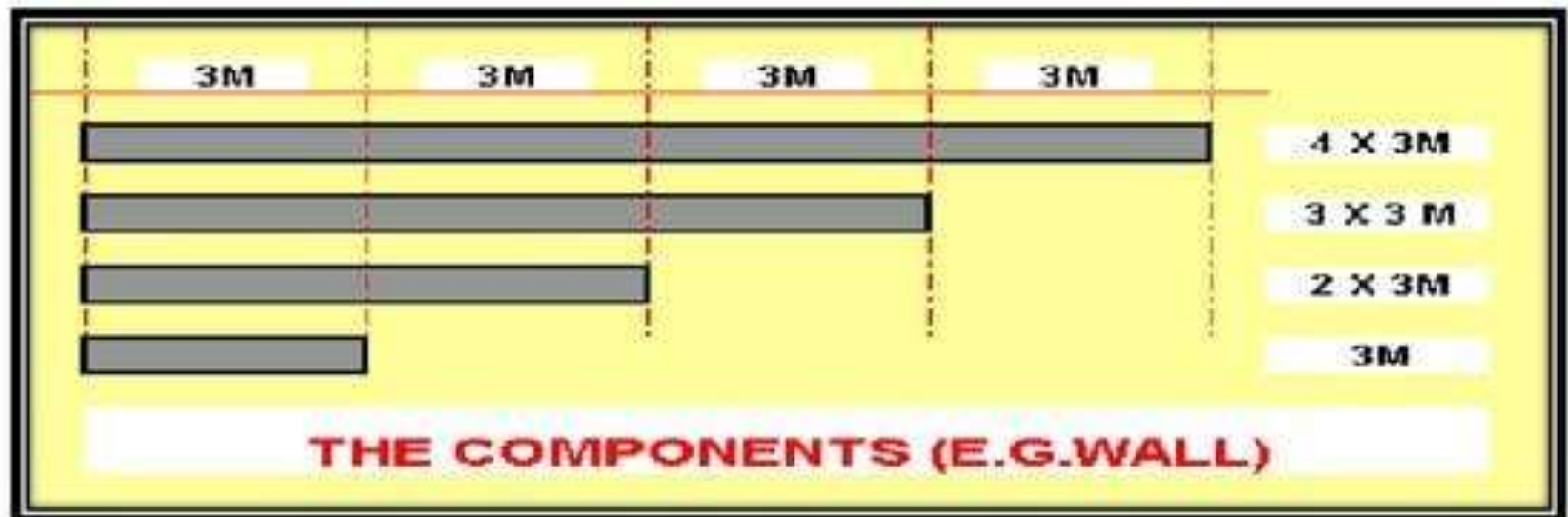
Modular Coordination

- Modular coordination is a concept of coordination of **dimension** and **space** in which building components are positioned in terms of **basic unit** or **MODULE(M)**
 - $1M = 100mm$
- It is internationally accepted by the **International Standard Organization** and many other countries including Malaysia.
- **A module:** a basic dimension which could for example form the basis of a planning **grid** in terms of **multiples** and **submultiples** of standard module.

Modular Coordination (Cont.,)

AIMS:

- To achieve dimensional compatibility between building dimensions, span, or spaces and the sizes of components or equipment by using related modular dimensions.
- Making the planning simpler & clearer by distinct indication of location of the building component in the building, both in respect to each other & a modular grid.
- Simplification of site work.
- Limiting the member of sizes of building component so that the linkage is based on modular measurement.



Modular Coordination (Cont.,)

Basics of module :

- The basic module is known as 1M which is equivalent to 100mm.

$$1M = 100mm$$

There are three type of MODULE :-

(I) Basic Module :

It is the **fundamental unit of size** in modular coordination and for general application to building & components. The size of basic Module is taken as **100mm denoted by "M"**.

(II) Multi Module :

multiples of basic module usually expressed in as "M" with numeric prefix as 2M , 3M, 4M etc are referred to as multi module.

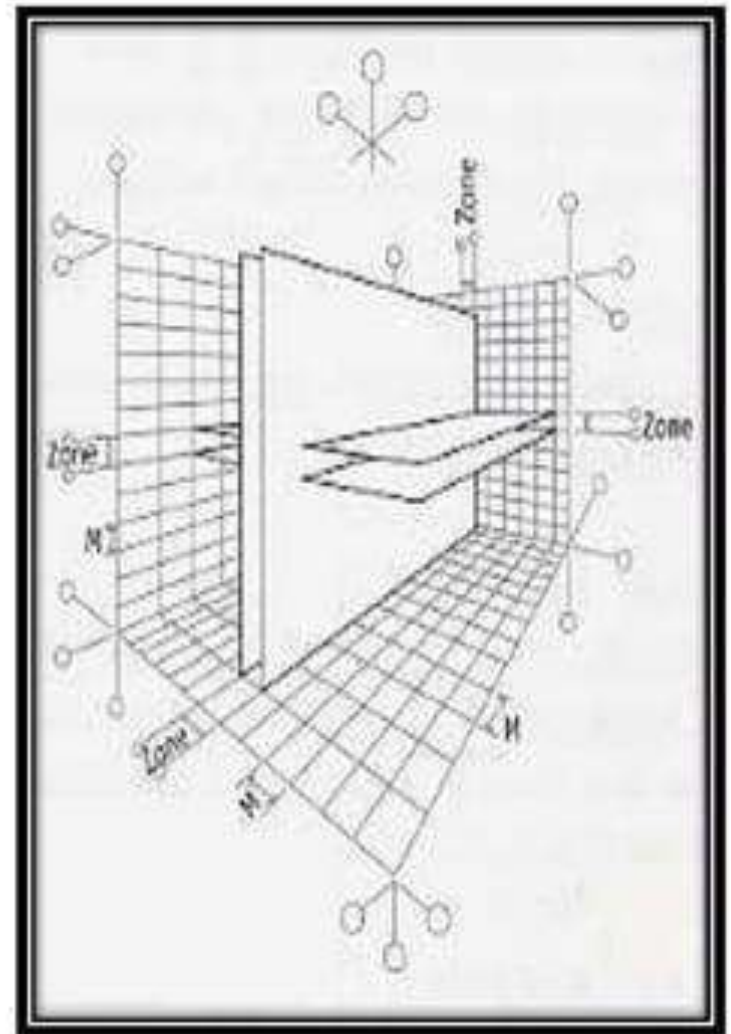
(III) Sub Module:

smaller than the basic module. For practical considerations, this sub modular increment shall be expressed as "M" with fractional prefix as 1/5M, 1/4M, 1/3M,etc.

Modular Coordination (Cont.,)

Modular Reference System

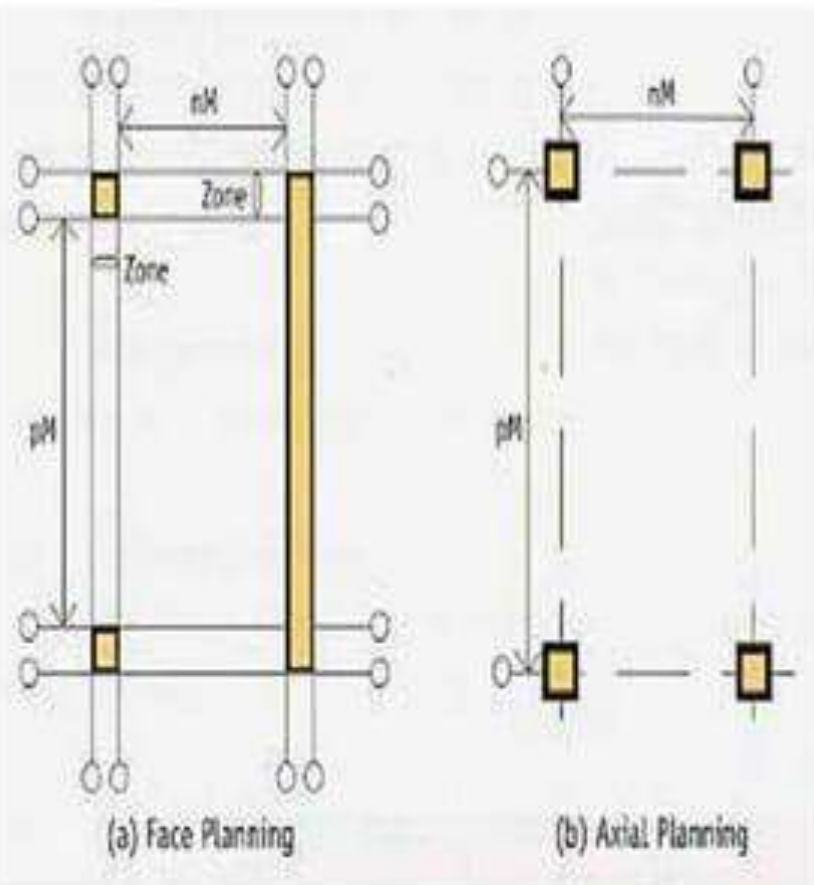
- The term modular reference systems is the three dimensional system of orthogonal space coordinates within the positions and sizes of components, elements and installations can be related by references to points, lines, or planes.



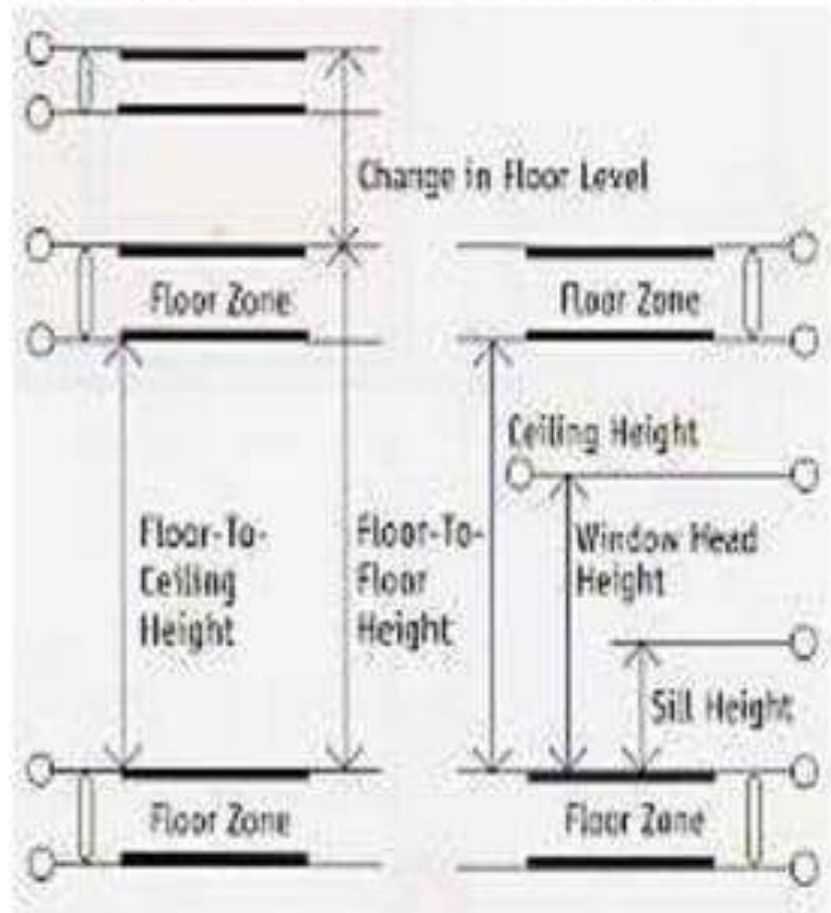
Modular Coordination (Cont.,)

CONTROLLING REFERENCE SYSTEM

Horizontal Controlling dimensions



Vertical Controlling dimensions

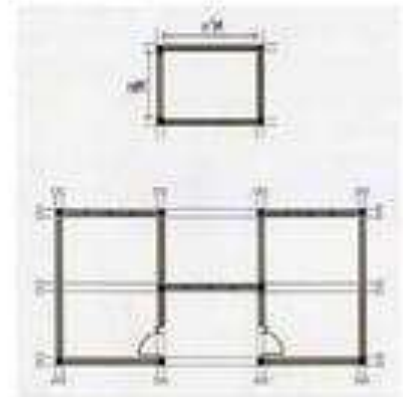
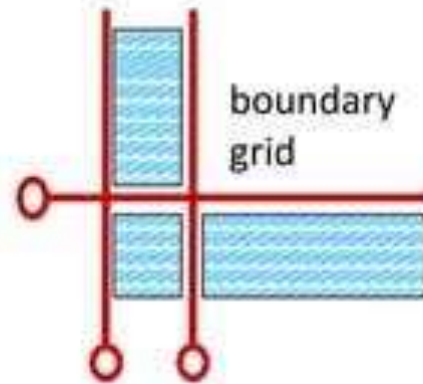


Modular Coordination (Cont.,)

Positioning of components and space

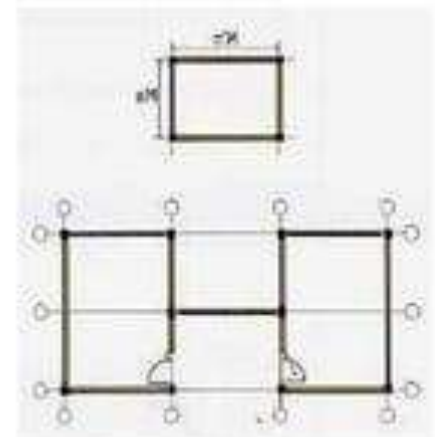
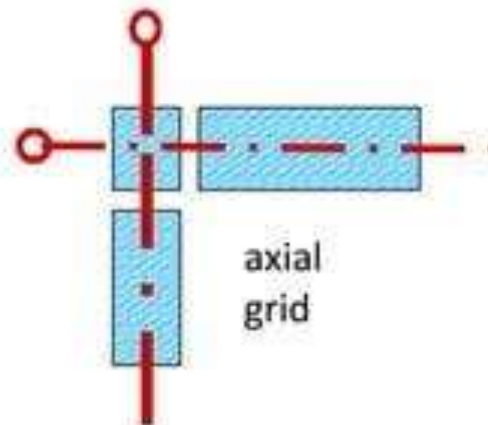
1. Boundary reference

- coordinates the position of the building components.
- determines the nominal size of components.
- placement of component within two Parallel modular Coordinating grids or planes so that it fills the space.



2. Axial reference

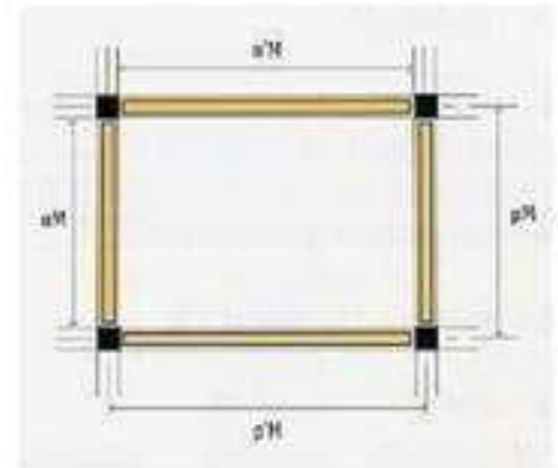
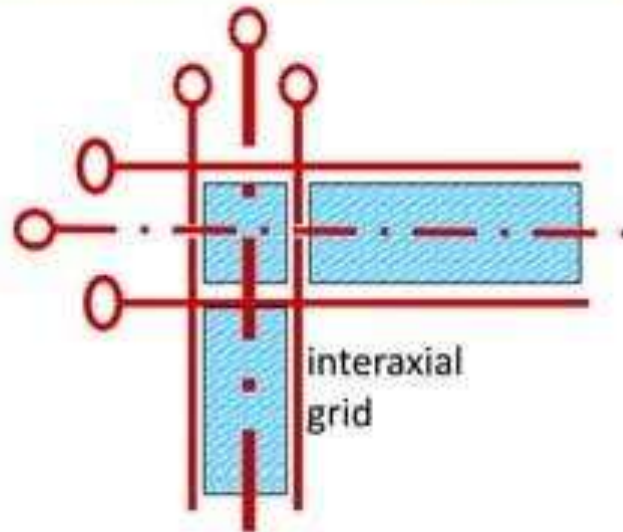
- coordinates the position of a components by placing the component so that the middle- axis coincides with a modular coordinating grid.



Modular Coordination (Cont.,)

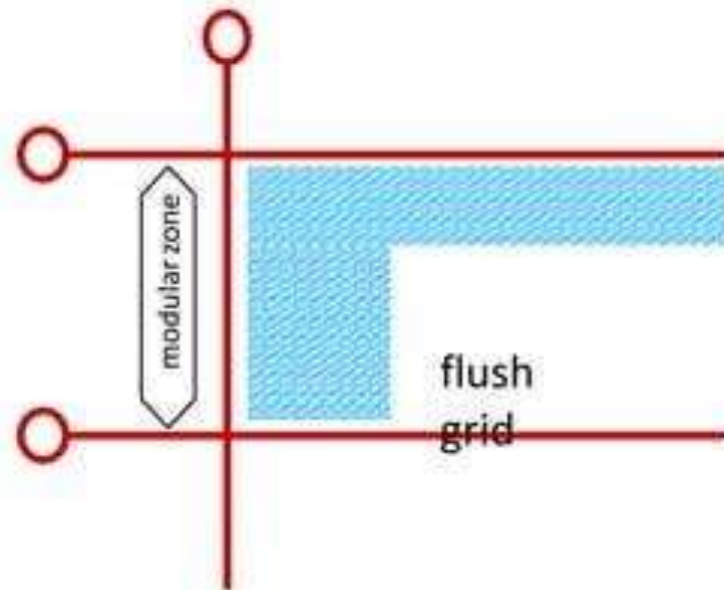
3. Interaxial reference

- coordinates the position and dimension of building component by a reference.



4. Flush reference

- coordinates the position of components by placing one surface of the component flush on to a modular coordinating grid or plane.



Modular Coordination (Cont.,)

Positioning of Building Components:

- **Structural components**

- Columns
- Beams
- Floor slabs
- Walls
- Staircases and lift cores

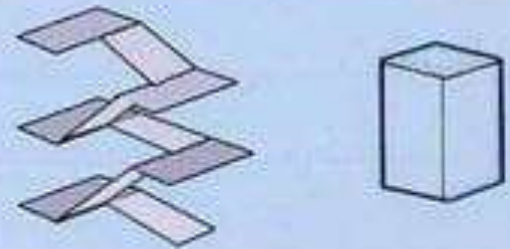
- **Non structural components**

- Doors
- Windows

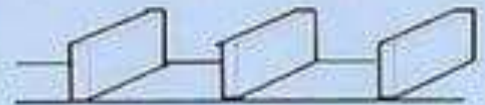
- **Finishes**

- Ceiling finishes
- Floor finishes
- Wall finishes

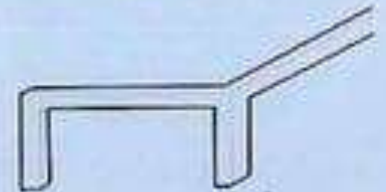
3 - D Component



Cross walls



Columns and beams (load bearing structures)



Floor Slab



Two methods of prefabrication

- Plant prefabrication
- Site prefabrication



Classification of Prefabrication

1. Small prefabrication
2. Medium Prefabrication
3. Large Prefabrication
 - Cast in Site Prefabrication
 - Off-Site (or) factory Prefabrication
4. Open system of prefabrication
5. Closed system of prefabrication
6. Partial prefabrication
7. Total prefabrication

Classification of Prefabrication (Cont.,)

1. Small prefabrication :

- For eg:- brick is a small unit precasted and used in buildings. This is called as small prefabrication. That the degree of precast element is very low.

2. Medium Prefabrication :

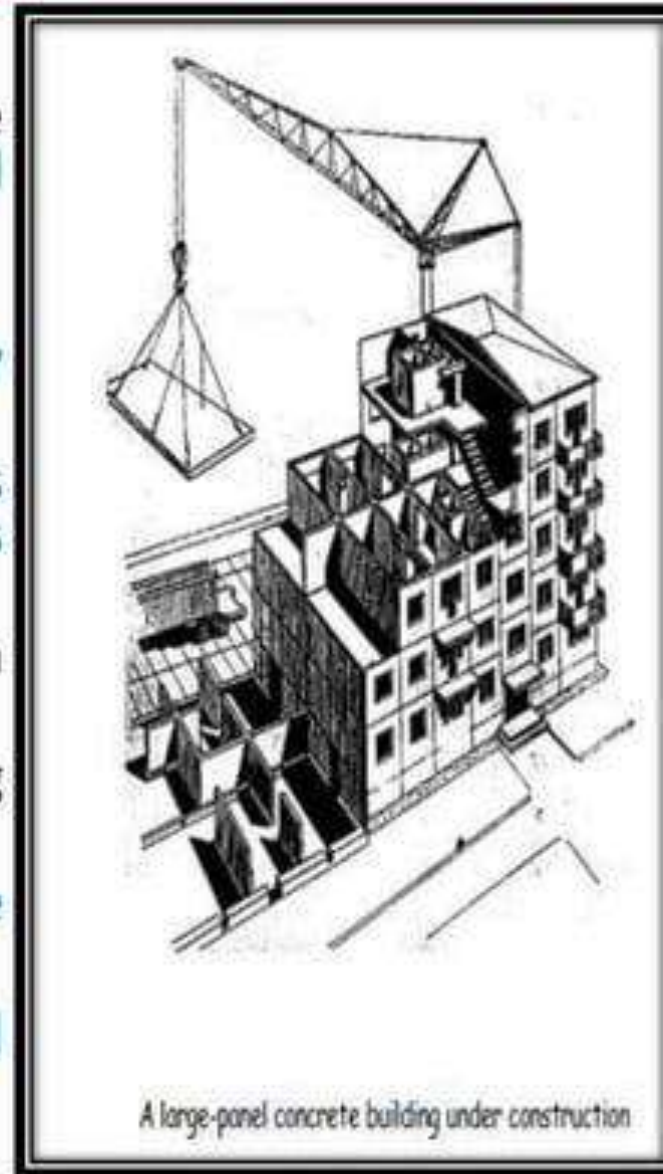
- Suppose the roofing systems and horizontal member are provided with precast elements.
- These constructions are known as medium prefabricated construction. Here the degree of precast elements are moderate.



Classification of Prefabrication (Cont.,)

3. Large Prefabrication:

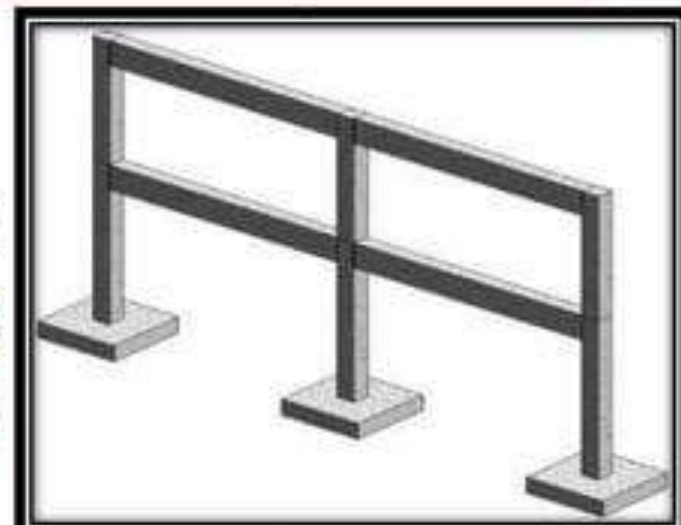
- In large prefabrication most of the members like wall panels, roofing/flooring systems, beams and columns are prefabricated.
- Here degree of precast elements are high.
- One of the main factors which affect the factory prefabrication is transport.
- The width of road and mode of transport vehicles are the factors which factor the prefabrications which is to be done off site (or) Factory.
- Small elements the conveyance is easier with normal type of lorry.
- On site prefabrication is preferred for following reasons:
 - factory situated at a long distance from the construction site.
 - vehicle have to cross a congested traffic.
 - heavy weighed elements to transport.



Classification of Prefabrication (Cont.,)

4. Open system of prefabrication:

- In the total prefabrication systems, the space framers are casted as a single unit and erected at the site.
- The wall fitting and other fixing are done on site.



5. Closed system of prefabrication :

- In this system the whole things are casted with fixings and erected on their position.



Classification of Prefabrication (Cont.,)

6. Partial prefabrication :

- In this method of construction the building element (mostly horizontal) required are precast and then erected.
- Since the costing of horizontal elements (roof / floor) often take there time due to erection of from work the completion of the building is delayed and hence this method is restored.
- Use of double tees, cored slabs, slabs etc., are some of the horizontal elements.
- This method is efficient when the elements are readily available when the building reached the roof level.
- The delay caused due to erection of formwork, delay due to removal eliminated completely in this method of construction Suitable for any type of building provided lifting and erection equipments are available.



Classification of Prefabrication (Cont.,)

7. Total Prefabrication :

- Very high speed can be achieved by using this method of construction.
- The choice of these two methods depend on the situations when the factory produced elements are transported and erected site we call it off-site prefabrication.
- If this method is to be adopted then we have a very good transportation of the products to site.
- If the elements are cast near the building site and erected, the transportation of elements can be eliminated, but we have to consider the space availability for establishing such facilities though it is temporary.
- The choice of the method of construction also depends on the following;
 1. Type of equipment available for erection and transport.
 2. Type of structural scheme (linear elements or panel)
 3. Type of connections between elements.
 4. Special equipment devised for special method construction.

Process (or) Production of Prefabricated Structures

Manufacturing process (on the production site)



Arranged the components before shipping

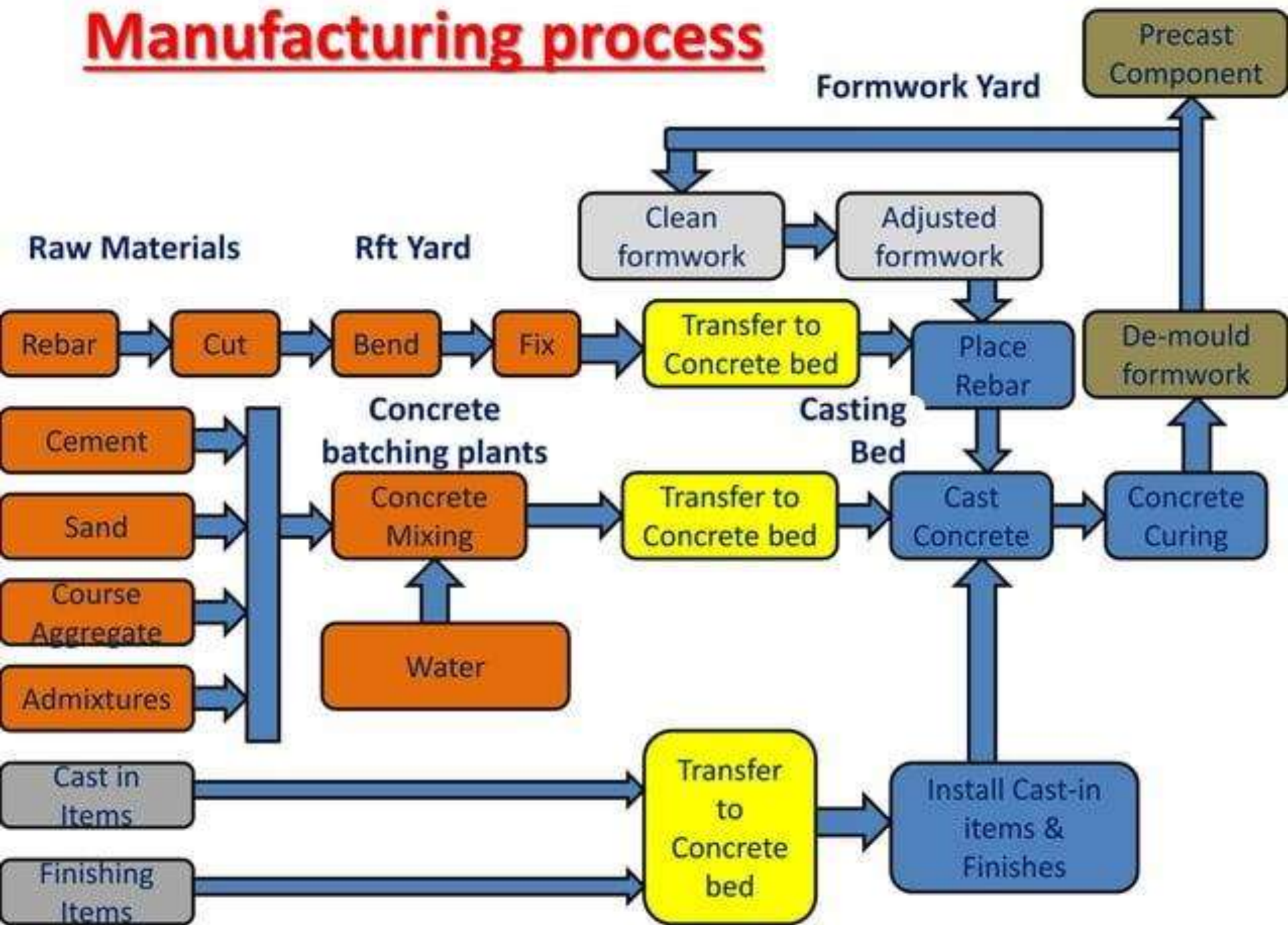


Transportation (to the construction site)



Erection (on the construction site)

Manufacturing process



STANDARDIZATION

- It is defined as the process of adoption of generally accepted uniform procedures, dimensions, materials or parts that directly affect the design prefabricated product or a facility.

ADVANTAGES OF STANDARDIZATION

- 1) Easier in design as it eliminates unnecessary choices
- 2) Easier in manufacture as there are limited number of variants.
- 3) Makes repeated use of specialized equipments in erection and completion
- 4) Easier and quicker.

STANDARDIZATION (Cont.,)

FACTORS INFLUENCING STANDARDIZATION:-

- To select the most rational type of member for each element from the point of production, assembly, serviceability and economy.
- To limit the number of types of elements and to use them in large quantities.
- To use the largest size of the extent possible, thus resulting in less number of joints.
- To limit the size and number of prefabricate by the weight in overall dimension that can be handled by the handling and erection equipment and by the limitation of transportation.
- To have all these prefabricates approximately of same weight very near to the lifting capacity of the equipment.

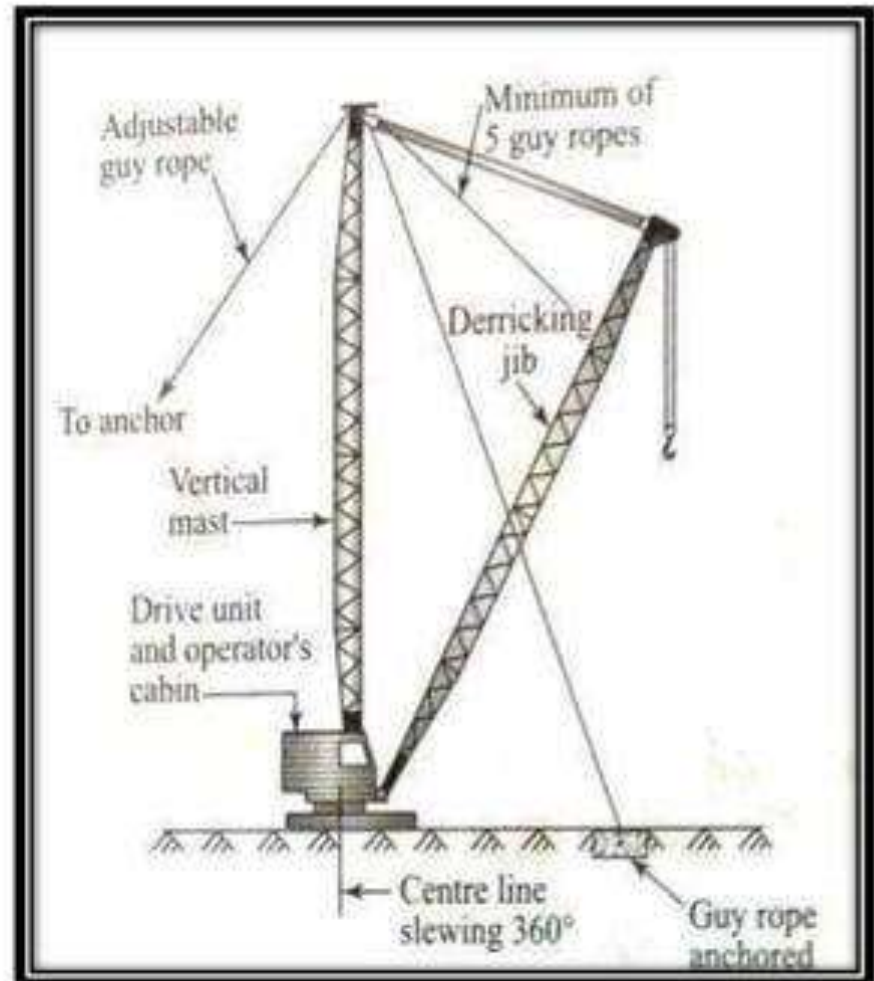
Types of cranes

- Stationary cranes (Fixed)
 1. Guyed derrick
 2. Climbing crane
 3. Tower crane
- Cranes on Rail (Movable)
 1. Gantry crane
 2. Tower crane
- Mobile Crane moving on ground
 1. Truck mounted
 2. Wheel Mtd Strut Boom Crane.

Stationary cranes (Fixed)

1. Gyped Derrick:

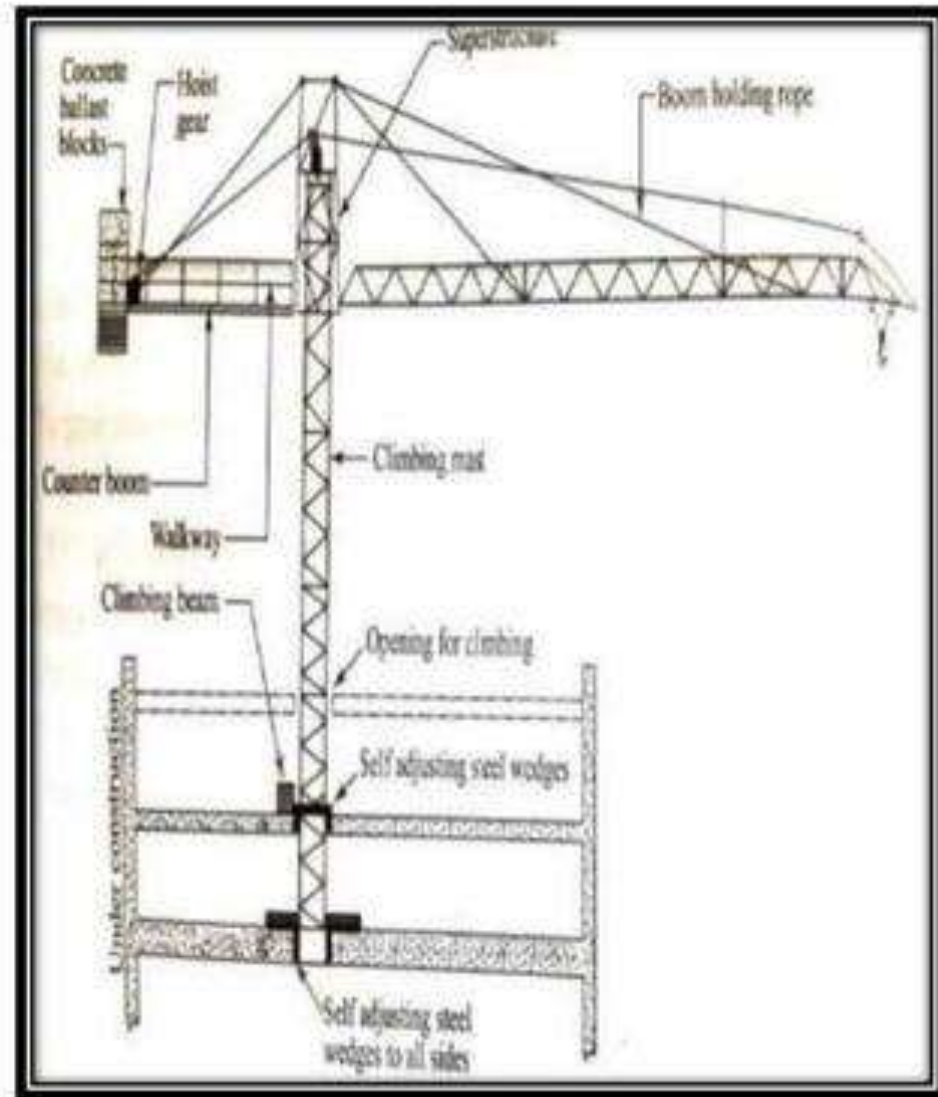
- Simple inexpensive non mobile equipment.
- Single lattice mast and jib.
- Mast stands vertical solid Bearing.
- At least five anchor ropes for Stability.
- Length is such to allow 360 deg swing.
- Used on framed buildings for erection of floor panel, columns .
- Lighter in weight.
- Can be shifted from floor to floor operating from an erected floor.



Stationary cranes (Fixed)

2. Climbing Tower Crane.

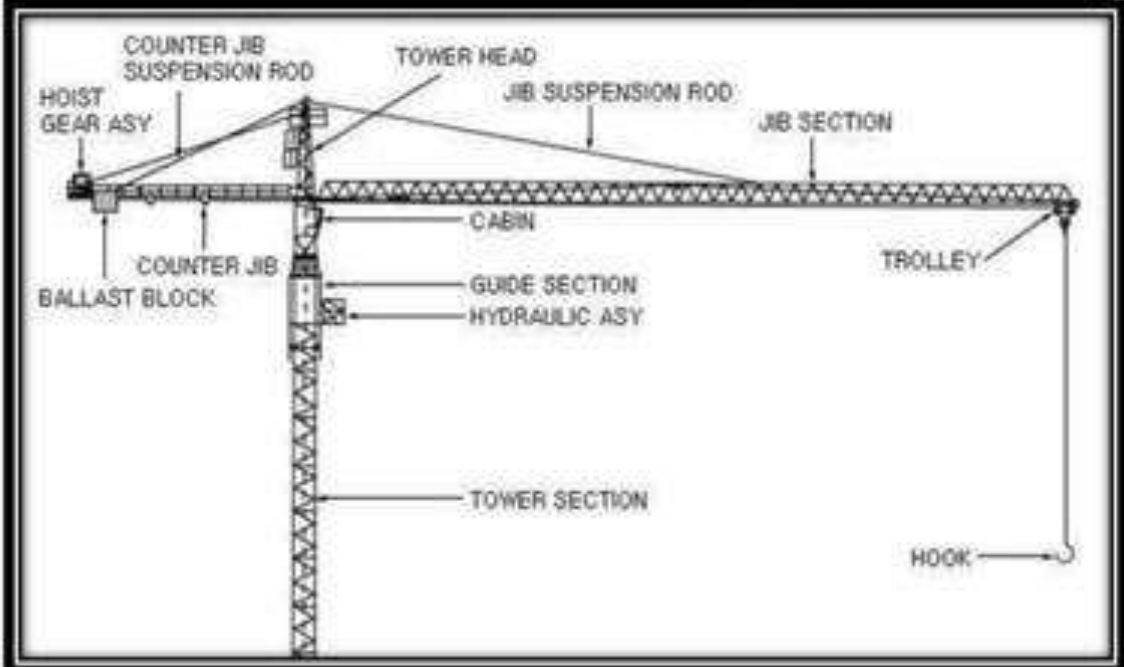
- Space not available.
- Tower crane is located inside.
- Up to 20 storeys constructed.
- Horizontal jib & balancing counterweight is placed on the top.
- Its operate 360 degree.
- Constructed in large area.



Stationary cranes (Fixed)

3. Tower Crane

- Balance crane fixed to the ground on a concrete slab.
- Height and lifting capacity are used in the construction of tall buildings.
- Three main parts: long horizontal jib, shorter counter-jib, and the operator's cab.
- Occupies less space with maximum output.
- Popular because of its power and efficiency.
- Fixed Tower Crane of up to 230 m height are available in India.



Cranes on Rail

1. Gantry Crane

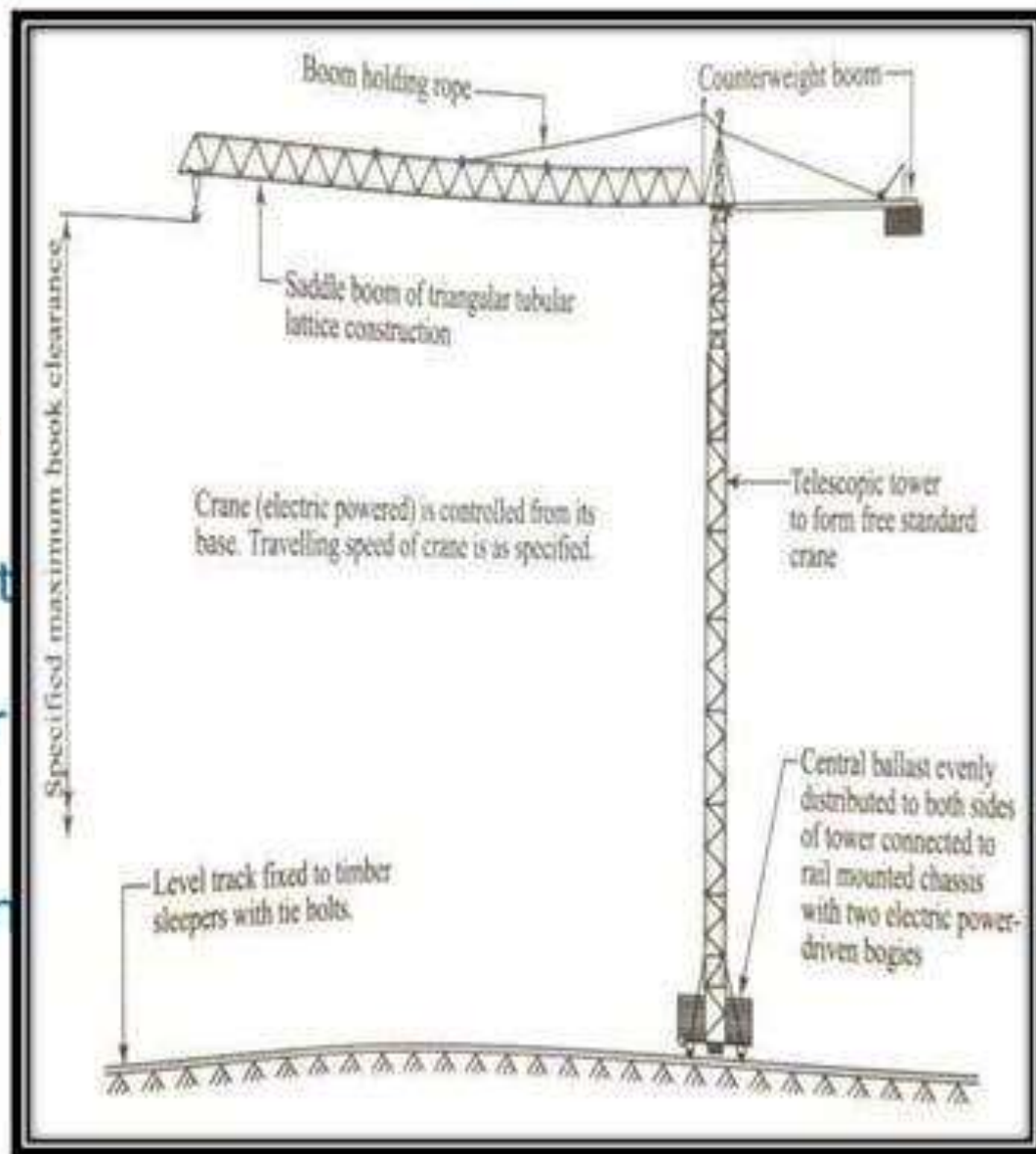
- Its used storage yards and in building where the travel is short.
- They consist two vertical leg and a one horizontal beam.
- A traveling cable winch attached to a wide beam stretching between rails/ wheels supported at high level.
- Advantage of this crane is more stable & handle heavy load.
- Disadvantages is no circular motion & longitudinal or horizontal movement is restricted.



Cranes on Rail

2. Rail Mounted Tower Crane

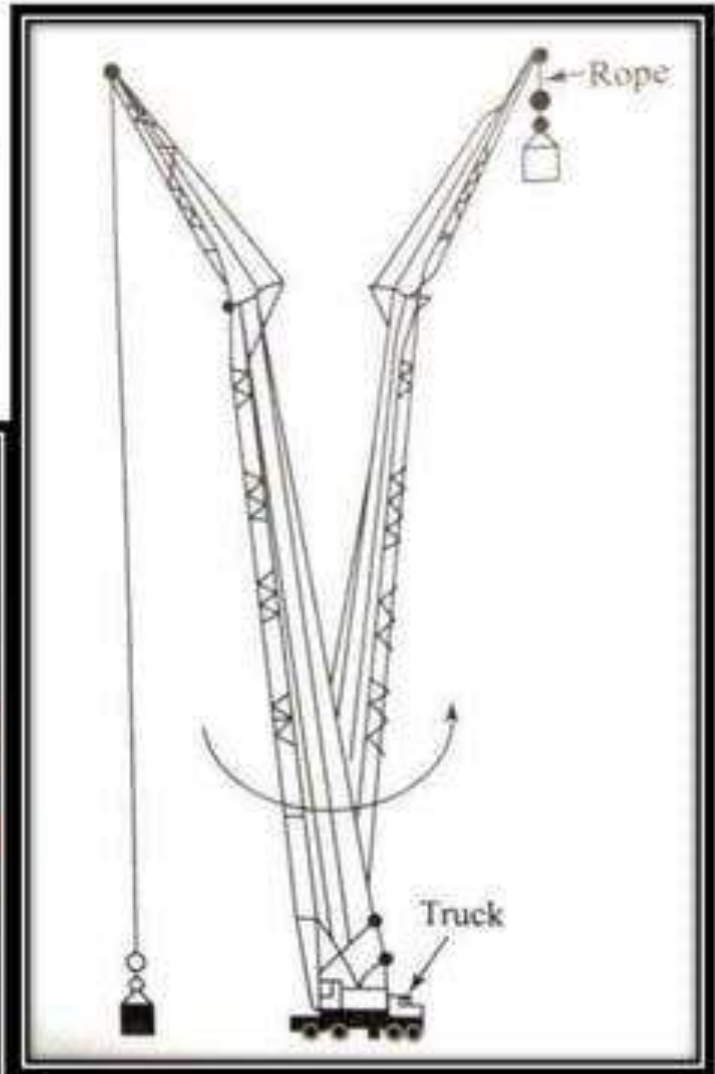
- Carries a load of one ton up to 16 ton.
- Heights up to 250 m.
- Optimum utilization of power.
- Loading of track will not affect the foundation.
- Central will have one or two operators cabin to facilitate view of operator at the erection height.



Mobile Crane moving on ground

1. Truck Mounted Tower Crane

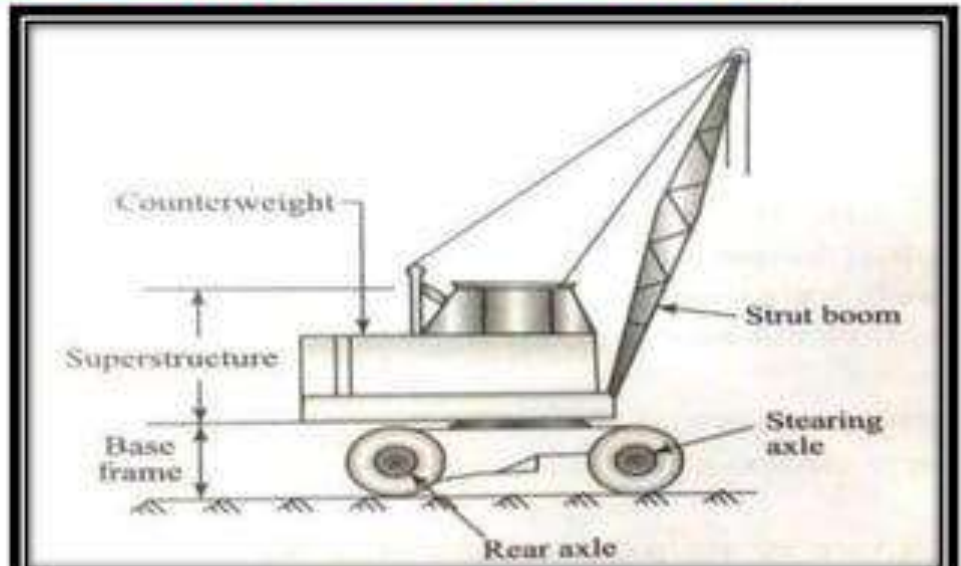
- Mounted on wheels.
- Lifting capacity of 100 tons or more.
- A longer boom is provided as fly jib is not used.



Mobile Crane moving on ground

2. Wheel Mtd Strut Boom Crane.

- Crane has three sections
 - Base frame.
 - Superstructure.
 - Jib.
- Turning capacity is 360 deg.
- Diesel engine powered.



Machineries used for handling



Stacking – Dos and Don't s

Horizontal stacking – beams , slabs and columns



Wall panels should always be stored vertically

